

Python Programming Lab

Lab 1 — Introduction, Print, Input & Functions

Beginner Level • Hands-On Practice

Covers: Environment Setup • print() in depth • input() • Variables & Data Types
Operators • Conditionals • Loops • Functions • Practice Exercises • Mini Project

Part 1: Environment Setup

1.1 Option A — Run Online (No Install)

Fastest way to start immediately. Use any of these in your browser:

- <https://replit.com>
- <https://www.online-python.com>
- <https://colab.research.google.com> (Google Colab, great for beginners)

1.2 Option B — Install Locally

1. Download Python 3.12+ from **python.org/downloads**
2. On Windows installer, check "**Add Python to PATH**"
3. Verify installation in terminal:

```
python --version
# Should print something like: Python 3.12.4
```

4. Install a code editor: **VS Code** (code.visualstudio.com) + Python extension

1.3 Running Your First File

```
python lab1.py
```

Save your code in a file named **lab1.py** and run it from the terminal using the command above.

Tip: Python files always end with **.py**. Save often and run after every small change — don't write 50 lines before testing.

Part 2: The print() Function — In Depth

print() is how Python displays output. It's more flexible than most beginners realize.

2.1 Basic Printing

```
print("Hello, World!")
print("Welcome to Python Lab 1")
print(123)
print(3.14)
print(True)
```

OUTPUT

```
Hello, World! Welcome to Python Lab 1 123 3.14 True
```

2.2 Printing Multiple Values

```
print("Name:", "Ravi", "Age:", 20)
print("A", "B", "C")
```

OUTPUT

```
Name: Ravi Age: 20 A B C
```

By default, print() separates values with a space and adds a newline at the end.

2.3 Customizing sep and end

```
print("apple", "banana", "cherry", sep=", ")
print("Hello", end=" ")
print("World")
print(1, 2, 3, sep=" -> ", end=" !!\n")
```

OUTPUT

```
apple, banana, cherry Hello World 1 -> 2 -> 3 !!
```

2.4 f-strings (Formatted Printing)

```
name = "Ravi"
age = 20
print(f"My name is {name} and I am {age} years old")
print(f"Next year I will be {age + 1}")

pi = 3.14159
print(f"Pi rounded: {pi:.2f}")
```

OUTPUT

```
My name is Ravi and I am 20 years old Next year I will be 21 Pi rounded: 3.14
```

2.5 Escape Characters in print()

```
print("She said, \"Python is fun!\")
print("Line1\nLine2\nLine3")
print("Column1\tColumn2\tColumn3")
```

OUTPUT

```
She said, "Python is fun!" Line1 Line2 Line3 Column1 Column2 Column3
```

Try It Yourself — print()

1. Print your name, roll number, and department on 3 separate lines using one print() with \n
2. Print "Python" 5 times on the same line separated by " - " using sep

3. Use an f-string to print: "The sum of 5 and 10 is 15" (calculate inside the f-string)

Part 3: The input() Function — In Depth

input() lets your program take data from the user. It ALWAYS returns a string.

3.1 Basic Input

```
user_name = input("What is your name? ")
print("Hello,", user_name)
```

SAMPLE RUN

```
What is your name? Ravi Hello, Ravi
```

3.2 Input Always Returns a String

```
age = input("Enter your age: ")
print(type(age))  # <class 'str'> -- even if user types 20
```

OUTPUT

```
Enter your age: 20 <class 'str'>
```

3.3 Type Conversion with Input (Very Important)

```
age = int(input("Enter your age: "))
height = float(input("Enter your height in meters: "))

print(f"Next year you'll be {age + 1}")
print(f"Your height in cm is {height * 100}")
```

SAMPLE RUN

```
Enter your age: 20 Enter your height in meters: 1.7 Next year you'll be 21 Your height in cm is 170.0
```

3.4 Taking Multiple Inputs

```
# Method 1: separate lines
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum:", a + b)

# Method 2: single line, split by space
x, y = input("Enter two numbers separated by space: ").split()
x = int(x)
y = int(y)
print("Product:", x * y)
```

SAMPLE RUN

```
Enter first number: 4 Enter second number: 5 Sum: 9 Enter two numbers separated by space: 3 6 Product: 18
```

3.5 Common Input Mistake (Type Error)

```
# WRONG -- this will crash
num = input("Enter a number: ")
print(num + 5)  # TypeError: can only concatenate str

# CORRECT
num = int(input("Enter a number: "))
print(num + 5)
```

Try It Yourself — input()

1. Ask the user for their name and city, then print "Ravi lives in Delhi" style sentence
2. Ask for two numbers and print their sum, difference, and product
3. Ask for radius of a circle and print its area (area = 3.14 * r * r)

Part 4: Variables and Data Types

```
name = "Ravi"           # str
age = 20                # int
height = 5.8            # float
is_student = True       # bool

print(name, age, height, is_student)
print(type(name), type(age), type(height), type(is_student))
```

OUTPUT

```
Ravi 20 5.8 True <class 'str'> <class 'int'> <class 'float'> <class 'bool'>
```

Type	Example	Description
str	"Ravi"	Text, in quotes
int	20	Whole numbers
float	5.8	Decimal numbers
bool	True / False	Boolean logic value

Part 5: Operators

5.1 Arithmetic Operators

```
a = 10
b = 3
print(a + b)    # 13  addition
print(a - b)    # 7   subtraction
print(a * b)    # 30  multiplication
print(a / b)    # 3.333... division (always float)
print(a // b)   # 3   floor division
print(a % b)    # 1   modulus (remainder)
print(a ** b)   # 1000 exponent (10^3)
```

5.2 Comparison Operators

```
print(5 == 5)   # True
print(5 != 3)   # True
print(5 > 3)    # True
print(5 < 3)    # False
print(5 >= 5)   # True
```

5.3 Logical Operators

```
age = 20
has_id = True
print(age >= 18 and has_id) # True
print(age < 18 or has_id)  # True
print(not has_id)          # False
```

Part 6: Conditional Statements

6.1 if / elif / else

```
marks = 75

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
else:
    print("Fail")
```

OUTPUT

```
Grade B
```

6.2 Nested Conditions

```
age = int(input("Enter age: "))
has_license = input("Do you have a license? (yes/no): ")

if age >= 18:
    if has_license == "yes":
        print("You can drive")
    else:
        print("Get a license first")
else:
    print("Too young to drive")
```

6.3 Even / Odd Check (Classic Example)

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print(f"{num} is Even")
else:
    print(f"{num} is Odd")
```

Part 7: Loops

7.1 for loop with range()

```
for i in range(1, 6):
    print("Count:", i)

# range(start, stop, step)
for i in range(0, 10, 2):
    print(i)
```

OUTPUT

```
Count: 1 Count: 2 Count: 3 Count: 4 Count: 5 0 2 4 6 8
```

7.2 while loop

```
n = 5
while n > 0:
    print(n)
    n -= 1
print("Blast off!")
```

7.3 Looping Through a String

```
word = "Python"
for letter in word:
    print(letter)
```

7.4 break and continue

```
for i in range(1, 10):
    if i == 5:
        break          # stop the loop entirely
    print(i)

for i in range(1, 6):
    if i == 3:
        continue       # skip just this iteration
    print(i)
```

7.5 Sum of First N Natural Numbers (Classic Example)

```
n = int(input("Enter n: "))
total = 0
for i in range(1, n + 1):
    total += i
print("Sum:", total)
```

Part 8: Lists (Basic Collections)

```
fruits = ["apple", "banana", "mango"]
print(fruits[0])      # apple
print(fruits[-1])     # mango (last item)

fruits.append("orange")
print(fruits)

for fruit in fruits:
    print(fruit)

print(len(fruits))    # number of items
```

OUTPUT

```
apple mango ['apple', 'banana', 'mango', 'orange'] apple banana mango orange 4
```

Part 9: Functions — In Depth

9.1 Basic Function

```
def greet(name):  
    return "Hello, " + name + "!"  
  
print(greet("Ravi"))  
print(greet("Simran"))
```

OUTPUT

```
Hello, Ravi! Hello, Simran!
```

9.2 Function with Multiple Parameters

```
def add(a, b):  
    return a + b  
  
result = add(5, 7)  
print(result)
```

9.3 Default Parameter Values

```
def greet(name, greeting="Hello"):  
    return f"{greeting}, {name}!"  
  
print(greet("Ravi"))           # Hello, Ravi!  
print(greet("Ravi", "Good morning")) # Good morning, Ravi!
```

9.4 Function Returning Multiple Values

```
def calculate(a, b):  
    return a + b, a - b, a * b  
  
s, d, p = calculate(10, 4)  
print("Sum:", s)  
print("Difference:", d)  
print("Product:", p)
```

OUTPUT

```
Sum: 14 Difference: 6 Product: 40
```

9.5 Function with No Return (just prints)

```
def print_table(n):  
    for i in range(1, 11):  
        print(f"{n} x {i} = {n*i}")  
  
print_table(5)
```

9.6 Function Calling Another Function

```
def square(n):  
    return n * n  
  
def sum_of_squares(a, b):  
    return square(a) + square(b)  
  
print(sum_of_squares(3, 4)) # 9 + 16 = 25
```

9.7 Function That Uses input() Internally

```
def get_user_age():  
    age = int(input("Enter your age: "))  
    return age  
  
age = get_user_age()  
if age >= 18:  
    print("Eligible to vote")  
else:  
    print("Not eligible yet")
```

Try It Yourself — Functions

1. Write a function **is_even(n)** that returns True/False
2. Write a function **max_of_three(a, b, c)** that returns the largest number
3. Write a function **factorial(n)** using a loop
4. Write a function **celsius_to_fahrenheit(c)** that returns $F = (C * 9/5) + 32$

Part 10: Mini Project — Putting It All Together

This program combines `print()`, `input()`, type conversion, conditionals, and functions.

```
def check_grade(marks):  
    if marks >= 90:  
        return "A"  
    elif marks >= 75:  
        return "B"  
    elif marks >= 60:  
        return "C"  
    else:  
        return "Fail"  
  
name = input("Enter your name: ")  
marks = int(input("Enter your marks: "))  
grade = check_grade(marks)  
  
print(f"\nStudent Report")  
print(f"Name   : {name}")  
print(f"Marks  : {marks}")  
print(f"Grade  : {grade}")
```

SAMPLE RUN

```
Enter your name: Ravi Enter your marks: 82 Student Report Name : Ravi Marks : 82 Grade : B
```

Part 11: Full Practice Exercise Set

A. print() Exercises

1. Print a simple ASCII box using only print() and \n
2. Print numbers 1 to 5 on one line separated by commas using sep
3. Print your full details using one single f-string

B. input() Exercises

1. Take 3 subject marks as input and print the average
2. Take principal, rate, and time as input, calculate simple interest
3. Take a temperature in Celsius and convert to Fahrenheit

C. Conditionals Exercises

1. Check if a number is positive, negative, or zero
2. Check if a year is a leap year
3. Simple calculator: take 2 numbers and an operator (+, -, *, /) as input

D. Loops Exercises

1. Print the multiplication table of a given number
2. Find the factorial of a number using a while loop
3. Print all even numbers between 1 and 50

E. Functions Exercises

1. Write a function to check if a number is prime
2. Write a function that reverses a string
3. Write a function that counts vowels in a word

Lab Report Tip: For each exercise, note down: (1) the code, (2) a screenshot of the output, and (3) one line on what concept it tested. This is what most instructors expect in a lab file.